SAHAI

सहाय

Voice-First AI Financial Companion for Bharat

Product Requirements Document

BITSoM × Masai — Product Management with GenAI & Agentic AI
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Product	SAHAI — Voice-First Financial AI Companion
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1. Executive Summary

1.1 Product Overview

SAHAI (सहाय) is a voice-first AI financial companion built for underserved Indian users in Tier 2 and Tier 3 cities. The name comes from the Hindi word for *help* or *assistant*, which is the product's core purpose. SAHAI lets users ask financial questions in their natural spoken language (Hinglish), get simple, actionable guidance in seconds, and receive proactive reminders on WhatsApp — without needing financial literacy, a new app download, or even the ability to type.

1.2 Product Branding and Visual Identity

- **Brand Name:** SAHAI (सहाय) AI
- **Strategic Rationale:** Derived from the Hindi word for Help/Assistant, the name strips away the intimidating, cold feel of "Artificial Intelligence" and replaces it with something approachable and local — a helping hand, not just a system.
- **Visual Asset Breakdown:**
  - *Interlocking Hands (Trust):* moving away from impersonal software toward a human partnership of equals built on mutual trust.
  - *Shield (Security):* communicates institutional-grade security, data protection and regulatory compliance.
  - *Growth Bars & Rupee Coin (Prosperity):* connect the technology directly to economic outcomes and affordable, data-driven financial access.
  - *Circuit Nodes at the Base (Technology):* show that human trust and financial growth are structurally supported by AI at the foundation.
- **Brand Palette:** Deep Terracotta Mud, Electric Teal, Rupee Gold.



Figure 1 — SAHAI brand identity and logo breakdown

1.3 Vision

> **Vision:** Every underserved Indian, regardless of financial literacy or digital fluency, should have access to a trusted, always-available financial companion who speaks their language and tells them exactly what to do.

> *"SAHAI is a voice-first AI financial guidance assistant that helps underserved Indians make better financial decisions through regional-language conversations and proactive WhatsApp nudges."*

SAHAI is not a fintech product. It is a **decision companion**. The goal is not to execute financial transactions — it is to give users the confidence to make the right call before they act. When users think about money, their first instinct should be: *"Let me ask SAHAI."*

1.4 Problem Statement

Over 500 million working-age Indians — auto drivers, daily wage workers, ASHA workers, small shopkeepers — make critical financial decisions every day without reliable guidance. They cannot afford financial advisors. They distrust banks. They are overwhelmed by complex fintech apps. The result is a cycle of poor financial decisions: unnecessary loans, missed government benefits, no savings buffer, and vulnerability to predatory lenders.

1.5 Proposed Solution

SAHAI replaces the need for financial literacy with conversational AI. Instead of teaching users about finance, it gives them a single clear answer in Hinglish: what to do, right now, given their situation. It works through voice (no typing required), WhatsApp (already installed on 500M+ Indian phones), and proactive nudges (so it reaches the user, not the other way around).

1.6 Methodology

Methodology: Agile (Iterative MVP Development). SAHAI was developed MVP-first, delivering and validating the core value proposition before expanding functionality.

Why Agile?

- Enables rapid prototyping and continuous improvement.
- Allows quick testing of AI, voice and WhatsApp features.
- Incorporates user feedback throughout development.
- Reduces risk by validating core assumptions before building advanced features.

2. Background & Context

2.1 The Financial Accessibility and Literacy Gap in India

India's financial inclusion story is incomplete. While Jan Dhan accounts reached 500 million users, account dormancy remains above 30%. Pradhan Mantri Suraksha Bima Yojana (PMSBY) covers less than 20% of eligible workers, and Mudra loan disbursals are heavily skewed toward urban, literate borrowers. **The gap is not access — it is guidance.** Users have bank accounts they don't use, insurance schemes they don't know about, and savings potential they can't unlock because no one explains these things in their language, at their level.

The credit picture is similar. India has a **₹28 lakh crore unorganized credit gap**. While **85% of smallholders have active bank accounts, less than 25% can access formal loans** because of complicated branch paperwork. Of the 7.7 crore (77 million) Udyam-registered enterprises in India, only 3.6 crore are credit-active across commercial or personal business loans — formal credit penetration of just 47%, leaving more than 4 crore registered entities untapped. SAHAI aims to act as the paperless, digital pipeline that closes this gap.

2.2 Challenges Faced by Underserved Users

- **Language barrier:** financial content is predominantly in English or formal Hindi.
- **Digital literacy gap:** most users cannot navigate complex apps or fill online forms.
- **Trust deficit:** users distrust banks and financial institutions because of past exploitation.
- **Time constraints:** daily wage earners cannot spend hours researching financial options.
- **Scheme awareness:** over 60% of eligible beneficiaries are unaware of applicable government schemes.
- **No personalization:** generic advice does not account for income volatility, family size or occupation.

2.3 Why Existing Solutions Fail

Solution	Why it fails for underserved users
Traditional banks	Require physical visits, literacy, complex paperwork
Fintech apps (Paytm, PhonePe)	Transaction-focused, not guidance-focused. No vernacular AI advice.
Government portals	Inaccessible UI, English-heavy, no personalization
WhatsApp forwards	Unverified, no personalization, no two-way interaction
Human advisors	Expensive, unavailable in rural/semi-urban areas, often incentivized to mis-sell

3. Target Users

3.1 Primary User Segments

- **Auto / Cab Drivers** — irregular income, high loan exposure, need EMI planning
- **ASHA / Anganwadi Workers** — low fixed stipend, eligible for multiple government schemes
- **Kirana / Small Shopkeepers** — mix of personal and business finance, Mudra loan eligible
- **Daily Wage Construction Workers** — weekly income, no savings habit, family dependents
- **Domestic Workers** — monthly income, unfamiliar with insurance and PF schemes
- **Small & Marginal Farmers** — seasonal income and lack of formal credit

3.2 User Personas

Personas 1–3 are the three demo personas implemented in the MVP. Persona 4 is a research persona that informs the post-MVP roadmap.

Persona 1 — Ramesh Kumar

Attribute	Detail	Attribute	Detail
Occupation	Auto Rickshaw Driver	**City**	Delhi NCR
Age	34 years	**Family**	Wife + 2 children
Income	₹15,000–20,000/month (variable)	**Language**	Hindi / Hinglish
Loan Status	₹80,000 vehicle loan (EMI: ₹3,200)	**Savings**	None (hand-to-mouth)
Key Pain	EMI timing vs irregular income	**Goal**	Build a ₹10,000 emergency fund
SAHAI Use	"Kya mujhe aaj EMI bhar deni chahiye?"	**Scheme Eligible**	PMSBY, PM Jan Dhan

Persona 2 — Priya Devi

Attribute	Detail	Attribute	Detail
Occupation	ASHA Community Health Worker	**Location**	Lucknow, UP
Age	29 years	**Family**	Husband + 3 children
Income	₹8,000–12,000/month (govt stipend)	**Language**	Hindi
Loan Status	No loan	**Savings**	₹500/month (inconsistent)
Key Pain	Unaware of applicable govt schemes	**Goal**	Secure children's education
SAHAI Use	"Mujhe kaunsi sarkari yojana milegi?"	**Scheme Eligible**	Sukanya Samriddhi, PMJJBY, Atal Pension Yojana

Persona 3 — Suresh Yadav

Attribute	Detail	Attribute	Detail
Occupation	Kirana Shop Owner	**Location**	Kanpur, UP
Age	42 years	**Family**	Wife + 2 children + parents
Income	₹30,000–40,000/month (variable)	**Language**	Hinglish
Loan Status	Mudra loan enquiry pending	**Savings**	₹3,000/month (ad hoc)
Key Pain	Should he take a business loan?	**Goal**	Expand shop inventory
SAHAI Use	"Business ke liye loan lena sahi hai kya?"	**Scheme Eligible**	PM Mudra Yojana, PMEGP

Persona 4 — Smallholder Farmer (Research Persona)

Attribute	Detail	Attribute	Detail
Occupation	Smallholder Farmer (Paddy)	**Location**	Shahganj, Jaunpur, UP (Tier 3 town)
Age	46 years	**Family**	Wife + 3 children + dependent brother
Income	₹15,000–25,000/month (seasonal)	**Language**	Hinglish / Awadhi
Loan Status	Kisan Credit Card (KCC) renewal pending	**Savings**	₹2,000/month (lumpy)
Key Pain	High-interest moneylender debt; irregular income; crop risk	**Goal**	Access institutional crop loans and reduce debt
SAHAI Use	"Kisan Credit Card ke liye apply kaise karein? Kaun se document chahiye?"	**Scheme Eligible**	Kisan Credit Card, PM Fasal Bima Yojana

4. Product Goals

4.1 Business Goals

Business Goal	Success Indicator
Demonstrate AI-first financial guidance	Show that LLM + voice can replace a human financial advisor for simple daily decisions
Increase scheme discovery	Surface relevant government schemes to users who are eligible but unaware
Drive habit formation via nudges	Use proactive WhatsApp messages to build consistent financial behaviour
Prove vernacular AI viability	Demonstrate that regional voice AI can serve Tier 2/3 users at scale
Establish trust through simplicity	Show that less complexity = more trust for this user segment

4.2 User Goals

User Goal	How SAHAI Delivers It
Get one clear answer	User asks a financial question and gets a single actionable response, not a list of options
Discover relevant government schemes	User learns about PM Jan Dhan, PMSBY, Mudra, Sukanya Samriddhi, etc. that apply to them
Build better financial habits	Regular nudges help the user avoid overspending, remember EMIs and save consistently
Feel understood	AI speaks in the user's language (Hinglish) and references their specific situation
Zero learning curve	User can use SAHAI on day 1 without any onboarding or training

5. Detailed Problem Statement

5.1 Current State

A daily wage worker in India earning ₹15,000/month faces at least 3–4 high-stakes financial decisions every month: when to pay EMIs given irregular income, whether to take a loan, which insurance to buy, and where to keep savings safely. Without guidance, they rely on advice from family (often equally uninformed), WhatsApp forwards (often misleading), or local moneylenders (often predatory).

5.2 Pain Points

- No trusted, affordable, always-available financial advisor
- Government scheme information buried in English-heavy portals with complex navigation
- Fintech apps designed for urban, literate, English-speaking users
- Financial decisions made under time pressure without data
- Loan-trap vulnerability — predatory informal lenders fill the guidance vacuum
- No proactive reminders — users forget EMI dates, savings commitments or renewal deadlines

5.3 Impact on Users

- ₹12,000+ in annual government scheme benefits missed (PMSBY, Jan Dhan, etc.)
- 30–40% higher effective interest rates from using informal credit instead of formal Mudra loans
- Zero emergency fund — one medical event wipes out 3 months of income
- No insurance coverage despite being eligible for PMSBY accident cover at just ₹20/year

5.4 The Opportunity

> **Opportunity:** A voice-first, WhatsApp-native, Hinglish AI companion can serve as the "trusted elder" this segment never had — available 24/7, consistent, unbiased and never pushy. The technology exists. The distribution channel (WhatsApp) exists. The gap is a product that connects them.

6. Market Sizing & STP

6.1 Segmentation, Targeting and Positioning

Persona	TAM (Total Addressable Market)	SAM (Serviceable Addressable Market)	SOM (Serviceable Obtainable Market) — Year 1 Target Pool
MSME Merchants *(Nano-retailers)*	63 million MSMEs in India	25 million merchants (unorganized/nano segment using digital payments)	**100,000 active merchants** (0.4% capture)
Farmers *(Small & marginal)*	140 million operational landholders	40 million farmers (digitally active, cash-crop belts, <2 hectares)	**60,000 active farmers** (0.15% capture)
SHGs of Women *(Micro-businesses)*	10 million NRLM-linked Self-Help Groups	3 million SHGs (enterprise-oriented, bank-linked micro-businesses)	**30,000 active SHGs** (1% capture via NGO partnerships)

Positioning: For underserved Indians in the unorganized sector who make daily money decisions without guidance, SAHAI is a voice-first financial companion that gives one clear, trusted answer in their own language — unlike banks, fintech apps and government portals that assume literacy, English and time.

7. The 7M Framework

The 7M framework describes SAHAI's long-term commercial model with regulated NBFC and bank partners. The current MVP is guidance-only (see Section 9, Scope).

7.1 Market — Our Target Segments

Focus: the digitally active unorganized sector (Month 12 target: 7,636 users)

- **MSME Merchants:** local kirana and retail shop owners.
- **Farmers:** smallholders in cash-crop belts.
- **SHG Women and ASHA welfare workers:** neighbourhood business collectives.

7.2 Mission — The Core Problem We Solve

Eliminating predatory lending through intelligent automation.

- Pull vulnerable segments out of high-interest informal debt traps.
- Automate complex online procedures for government schemes and loans.
- Provide simple, voice-guided investment guidance for group wealth.

7.3 Message — The "Trust" Hook

Simple, transparent, jargon-free communication.

- **For Merchants/Farmers:** "Your daily QR scans and land records are your credit history. Stop paying the local moneylender."
- **For SHG Women:** "Pool your small change in the Digital Gullak. Let your group's savings grow safely."

7.4 Media — Lean Sourcing Channels

- **For MSMEs & Farmers:** direct on-the-ground onboarding backed by the reputation of partner NBFC field officers, so users share profile details in a trusted, face-to-face setting.
- **For SHG Women:** in-person training circles where partner NBFC field staff guide the group through using the Digital Gullak feature.

7.5 Money — Revenue Generation (Year 1: ₹50.4 Lakh)

Transparent fee structures synced with the revenue model.

- **MSME:** flat ₹150 platform fee per drawdown (no compound interest).
- **Farmers:** 2% sourcing fee (₹500) paid by the lending bank per loan.
- **SHGs:** 0.5% annual AUM fee from fund managers for micro-savings.

7.6 Movement — The Mechanism & Flow

Frictionless onboarding and automated repayment cycles.

- **Onboarding:** single-OTP data ingestion via Account Aggregator.
- **Merchant repayment:** automated daily evening UPI auto-debits, matched to sales.
- **Farmer repayment:** seasonal, harvest-synced repayments.

- **SHG savings:** "Digital Gullak" auto-rounds up change into fractional funds.

7.7 Measurement — Success Metrics

Tracking the health and growth of the platform: funnel health (onboarding conversion at the OTP step) and stickiness ("repeat velocity" — how quickly users borrow again).

Metric	Target	Definition	Impact
Customer Repeat Rate	60% repeat borrowing within 60 days	% of users who take a new round of capital after successfully repaying their previous loan cycle	Proves deep adoption and doubles platform-fee revenue per counter with no new marketing spend
Sign-Up Funnel Completion	75% onboarding success	% of users who complete registration via the single-OTP bank-linkage step without dropping off	Proves the voice-guided, document-free design removes the friction that makes users abandon banking apps
Portfolio Default Rate (DPD 30)	<3.5%	Share of outstanding loan value more than 30 days past due	Keeps the portfolio in line with safe Indian micro-lending standards so partner NBFCs keep funding
Daily Collection Efficiency Index (CEI)	95% on-time daily repayment	% of automated evening UPI auto-debits that collect on the first attempt	Collects small payments exactly when shopkeepers have fresh cash from daily sales

8. Product Launch Strategy

SAHAI will launch as India's first AI-powered financial guidance network for underserved communities — directly through WhatsApp, local-language voice assistants and on-ground trust partners. Instead of asking users to learn finance, SAHAI explains benefits, loans and schemes in simple local language and helps them act immediately. Through **Financial Seva Camps**, **community ambassadors** and the **"Bank Bids for You"** campaign, SAHAI turns financial access from confusing and intimidating into simple, trusted, actionable guidance for every household.

8.1 WhatsApp-First Launch

Users don't need to install another app. We launch through:

- WhatsApp voice assistant
- Missed-call onboarding
- QR posters at kirana stores, banks, CSC centres and mandis

> **Example:** "Give a missed call and know which government benefits you can claim today." This makes onboarding accessible even for low-literacy users.

8.2 Community Trust Ambassadors

We partner with:

- SHG leaders
- Local teachers
- Kirana owners
- NGO workers
- MFI agents

They become "SAHAI Saathis" who:

- Educate users
- Assist onboarding
- Earn referral rewards

Trust spreads faster through familiar community faces than through digital ads.

8.3 Voice-Based Financial Stories

Short regional-language reels and IVR stories, such as *"How Ramesh saved ₹3,000/month"* and *"How Sunita got a subsidy through SAHAI"*, distributed through WhatsApp forwards, YouTube Shorts, community radio and loudspeaker announcements in local markets. This matches the behavioural insight from our research that people rely heavily on word-of-mouth and informal advice.

9. Scope

9.1 In Scope (MVP)

Feature	Implementation Note
Voice input via microphone	Browser MediaRecorder API → Sarvam AI STT
Text input fallback	Direct text field in UI
Hinglish AI responses	Groq Llama 3.3 70B with a custom system prompt
Text-to-speech output	Sarvam AI TTS (bulbul:v3)
WhatsApp nudges	Twilio WhatsApp Sandbox → recipient's phone
Incoming WhatsApp messages	Twilio webhook (`/whatsapp`) — user sends text or voice note, SAHAI replies
Persona-based memory	3 hardcoded demo personas; chat history in SQLite
Government scheme recommendations	Embedded in LLM system-prompt knowledge
Persona selection UI	Persona selector in React frontend
Nudge trigger panel	Button in UI → WhatsApp message sent instantly

9.2 Out of Scope (MVP)

Out of Scope	Reason
Smart profile linking / automatic financial verification	Requires RBI licensing and the Account Aggregator framework
Real financial transactions	Outside demo scope — guidance only
Credit scoring	Requires bureau integration (CIBIL/Experian)
Investment execution	SEBI compliance required
User authentication / login	Not needed for demo
Languages beyond Hindi/Hinglish	Phase 2 roadmap item
Production-grade security	MVP demo context only

10. Design Thinking Process

10.1 Internal Development Strategy

SAHAI is developed primarily in-house to keep control over the user experience and to iterate quickly on AI workflows, voice interactions and financial guidance. External services are used only for specialized infrastructure.

Why build internally?

- Faster product iteration
- Better control over UX
- Lower coordination overhead
- Easier AI experimentation
- Reduced vendor dependency

10.2 External Services Used

- Sarvam AI — speech (STT + TTS)
- Groq — LLM inference
- Twilio — WhatsApp messaging

10.3 Design Thinking

Stage 1: Empathize — Stepping into Their Worlds

Through contextual inquiries and in-depth interviews, we uncovered the real behaviours behind stated needs:

Segment	Stated Need	Hidden Iceberg
MSME Merchants	Hassle-free access to government schemes	Intense anxiety about bureaucratic procedures. They believe government portals are deliberately complicated or need middlemen, and they cannot leave their shops to visit administrative offices.
Farmers	Low-cost formal credit to avoid local moneylenders	Informal lenders hand over cash within 10 minutes at the farm gate with no paperwork. Banks need land records (7/12 extract) and take weeks. Farmers fear the social stigma of losing assets if a seasonal crop fails.
SHGs of Women	Group business funding and collaborative investment options	Technology access is unequal — smartphones are often shared among family members. Financial decisions need total transparency so the whole group agrees on how money is used.

Stage 2: Define — Framing the Root-Cause Problems

We turned these insights into actionable How Might We (HMW) questions:

- **MSMEs:** How might we simplify and automate finding and applying for government business schemes for busy MSME owners, given their worries about bureaucratic platforms and complex eligibility rules?
- **Farmers:** How might we match the speed of informal lenders while providing low-cost formal credit, even without digital documents and with seasonal income swings?
- **SHGs:** How might we create a transparent group underwriting and investment system for women's collectives, despite limited smartphone ownership and the need for shared financial goals?

Stage 3: Ideate — Exploring Possibilities via App Flexibility

The application has an adaptable core. At launch, users choose Merchant, Farmer or SHG, and that choice reshapes the backend workflows and interface:

- **MSME "Scheme Matchmaker":** an in-app AI document scanner. Merchants photograph their trade licence (Udyam Registration); OCR extracts the data, secure APIs connect to government systems, ~80% of the application is pre-filled automatically, and dense jargon is simplified into a one-page summary.
- **Farmer "Seasonal Cash Flow" System:** removes rigid monthly EMIs. It links to digital state land records (Bhulekh) via voice search to verify holdings instantly, and uses alternative data (satellite crop data, weather forecasts) to build a flexible schedule aligned with actual harvest seasons.
- **SHG "Group-Trust Ledger":** the "Digital Gullak" (shared pot) interface tracks the group's transaction history to calculate a safe daily saving amount and routes small sums into secure fractional investments. To handle shared devices, it runs in a "Single-Device Group Mode" that plays localized voice scripts aloud and requires multi-user voice approvals during meetings.

Stage 4: Prototype — Executing a Simple Voice-First MVP

To fit the technical limits of an early-stage MVP, the prototype skipped heavy engineering, complex data connections and document uploads. It focused instead on the most important layer: a simple, localized, voice-guided conversation.

- **Scope:** an interactive layout that simulates a conversational voice assistant speaking to the user in their dialect.
- **Persona adaptation:** the assistant adjusts its vocabulary to the selected persona's financial problem — low-interest institutional loans such as Kisan Credit Card for a farmer, government cash-flow schemes such as Mudra for a merchant, and how to set up a Digital Gullak savings pool for SHG women.
- **Launch boundary:** no file uploads or document scanning in the MVP. The system shows users which schemes apply to them and where to find affordable capital. Documentation and Account Aggregator pipelines are deferred to later stages.

11. User Journey

11.1 Primary Journey — Voice Financial Query

Stage	User Goal	What SAHAI Does	Channel	Likely Pain Point
Awareness	Understand what SAHAI is	Shares a clear promise: financial help in voice	WhatsApp, referral, voice	Low trust — "Is this for me?"
First Use	Try without effort	Greets in Hinglish and asks one simple question	Voice	Fear of complex setup
Problem Sharing	Explain a financial need	Listens and classifies the request	Voice	User may not know financial terms
Guidance	Get a useful answer	Gives a short recommendation and the reason	Voice + text	Advice must feel practical, not generic
Follow-Up	Remember what to do	Sends a summary, reminder or nudge	WhatsApp	Message fatigue if too frequent
Habit Building	Use again when needed	Uses memory and consistent rules	Voice + WhatsApp	Relevance must stay high
Expansion	Explore schemes or deeper help	Adds scheme recommendations and more personas over time	Voice + WhatsApp	Too many options can overwhelm

12. User Stories

12.1 Voice Guidance

> **US-01:** As Ramesh (auto driver), I want to speak my financial question in Hindi so that I don't have to type or know English to get advice.

> **US-02:** As a user, I want to hear the AI response spoken aloud in Hinglish so that I can understand it even if I struggle with reading.

12.2 Government Scheme Discovery

> **US-03:** As Priya (ASHA worker), I want to know which government schemes I am eligible for based on my occupation and income so that I don't miss out on benefits I deserve.

> **US-04:** As a user, I want SAHAI to explain how to apply for a scheme in simple steps so that I can actually complete the application.

12.3 Proactive Nudges

> **US-05:** As Ramesh, I want a WhatsApp reminder 2 days before my EMI due date so that I never miss a payment and avoid penalties.

> **US-06:** As Suresh (shopkeeper), I want weekly nudges about my spending vs savings so that I build better financial habits over time.

12.4 Persona-Aware Responses

> **US-07:** As a user, I want SAHAI to remember that I am an auto driver with a ₹3,200 EMI so that its advice is relevant to my actual situation, not generic.

13. Functional Requirements

FR-ID	Feature	Requirement	Priority	Status
FR-01	Voice Input	System must capture audio from the microphone via the browser MediaRecorder API and send it as a blob to the backend	**Must Have**	✓ Built
FR-02	Speech-to-Text	System must transcribe Hindi/Hinglish audio using Sarvam AI (saarika:v2.5) with <3s latency	**Must Have**	✓ Built
FR-03	LLM Response Generation	System must generate contextual Hinglish financial guidance using Groq Llama 3.3 70B with persona memory injected into the system prompt	**Must Have**	✓ Built
FR-04	Text-to-Speech Output	System must convert the LLM response to natural Hinglish audio using Sarvam AI (bulbul:v3) and auto-play it in the UI	**Must Have**	✓ Built
FR-05	WhatsApp Nudges	System must send proactive Hinglish messages via Twilio WhatsApp Sandbox to the registered recipient	**Must Have**	✓ Built
FR-06	Persona Selection	UI must allow selection of 3 demo personas; backend must load the corresponding memory profile	**Must Have**	✓ Built
FR-07	Text Fallback Input	User must be able to type a question when voice is unavailable	Should Have	✓ Built
FR-08	Transcript Display	System must display the STT transcript alongside the AI response for transparency	Should Have	✓ Built
FR-09	Scheme Recommendation	LLM must recommend relevant government schemes based on the user persona (occupation, income, family)	**Must Have**	✓ Built
FR-10	Nudge Trigger via UI	UI must include a button to manually trigger a WhatsApp nudge to the demo recipient number	**Must Have**	✓ Built
FR-11	Conversation Memory	System must store and retrieve the user's conversation history and profile data from SQLite per persona	Should Have	✓ Built
FR-12	CORS & API Security	Backend must load all API keys from environment variables and never expose keys in frontend code	**Must Have**	✓ Built

14. Non-Functional Requirements

NFR Category	Requirement
Response Time	End-to-end voice query response (STT + LLM + TTS) must complete within 5 seconds under normal network conditions
Availability	Backend should be deployable on a free-tier host (e.g. Railway) with >95% uptime during the demo period

NFR Category	Requirement
Language Support	System must handle Hinglish (Hindi + English mixed) and vernacular input and output natively without degraded accuracy
Mobile UI	Frontend must render correctly at 390px width (mobile-first), centred on desktop
API Key Security	All third-party API keys must be loaded via environment variables; no keys in source code or frontend bundles
Scalability	Architecture must support adding new personas or languages by modifying `personas.py` and the LLM system prompt only
Audio Format	STT must accept WAV/WebM from the browser; TTS must output base64 audio playable in a browser audio element
Error Handling	All API calls must have try/catch with user-facing fallback messages in Hinglish
Free-Tier Compliance	System must operate within the Groq, Sarvam AI and Twilio free/trial tier limits for the demo

15. Feature Prioritization (MoSCoW)

MUST HAVE	SHOULD HAVE	COULD HAVE	WON'T HAVE (MVP)
• Voice input (mic) • Hinglish STT (Sarvam) • LLM guidance (Groq) • Hinglish TTS (Sarvam) • Persona selection • WhatsApp nudges • Scheme recommendations • SQLite memory	• Text input fallback • Transcript display • Conversation history • Nudge preview in UI • Error messages in Hinglish • Audio response autoplay	• More language support • SMS nudges (MSG91) • Chat history export • Spending charts • Offline mode	• Bank account integration • Real UPI transactions • Credit score access • Investment execution • User login / auth • Production hardening

16. System Architecture

16.1 High-Level Architecture

SAHAI follows a clean 3-layer architecture: a React frontend (client), a FastAPI backend (server), and external AI/messaging APIs (services). All communication with external services is over HTTPS.

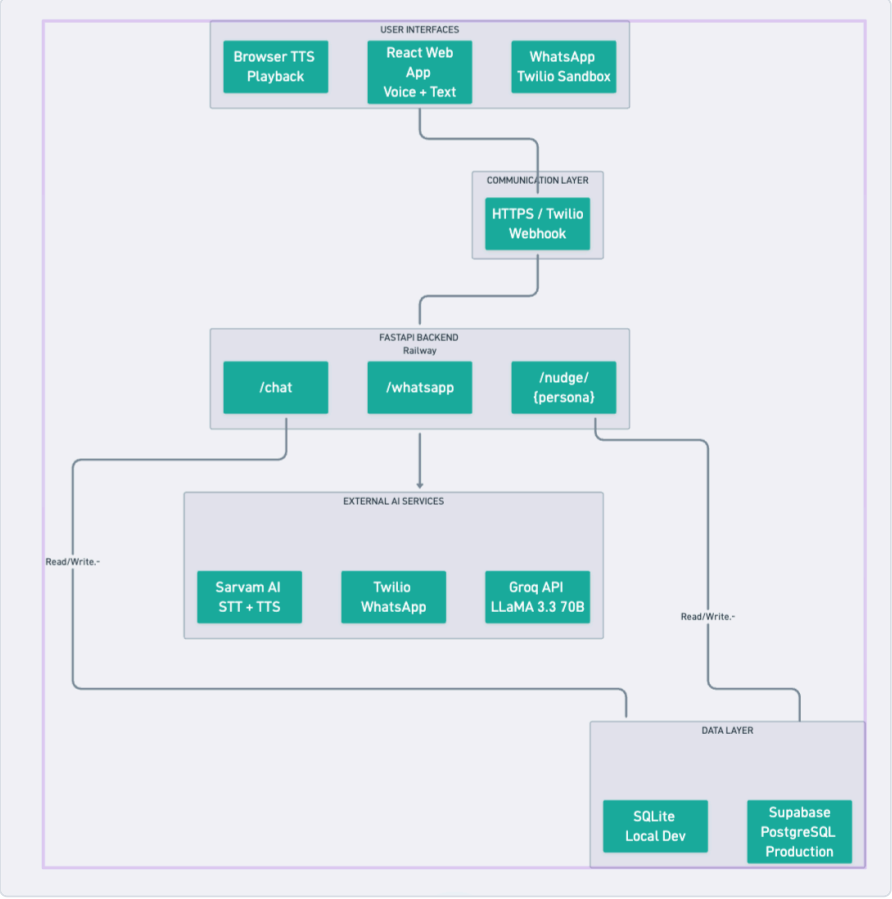


Figure 2 — SAHAI high-level system architecture

16.2 Request Flow — Voice Query

Step	Action
1. Capture	Browser MediaRecorder captures audio → sends blob to `POST /chat`
2. STT	FastAPI sends audio to Sarvam AI (saarika:v2.5) → receives transcript text
3. Memory	Backend fetches the persona profile and recent chat history (income, occupation, past queries)
4. Prompt Build	Transcript + user profile injected into the SAHAI system prompt
5. LLM Call	Groq Llama 3.3 70B generates a short Hinglish response
6. TTS	Response text sent to Sarvam AI (bulbul:v3) → receives base64 audio
7. Response	FastAPI returns `{ user_text, response_text, audio_base64 }`
8. Playback	React decodes base64 → plays audio and shows text on screen

16.3 Channel Strategy

- **Primary channel:** voice-first conversational interface
- **Engagement channel:** WhatsApp
- **Future channels:** IVR calls, SMS alerts, bank integrations

> **Why this strategy?** Users already trust and actively use WhatsApp, which reduces adoption friction while enabling continuous engagement.

16.4 Channel Parity Strategy

- Shared AI recommendation engine
- Unified user profile
- Consistent financial guidance
- Centralized scheme database
- Common business rules across channels

17. Technology Stack

Layer	Technology
Frontend	React 18 + Vite, Tailwind CSS
Backend	Python FastAPI (uvicorn, httpx)
Speech-to-Text	Sarvam AI — saarika:v2.5
Text-to-Speech	Sarvam AI — bulbul:v3
LLM	Groq Cloud — llama-3.3-70b-versatile
WhatsApp	Twilio WhatsApp Sandbox
Database	SQLite (MVP) → Supabase PostgreSQL (production)
Hosting — Frontend	Vercel (planned)
Hosting — Backend	Railway (planned)

18. Government Scheme Knowledge Base

SAHAI's LLM system prompt includes embedded knowledge of the following government schemes, mapped to user personas by eligibility criteria.

Scheme	Benefit	Eligibility	Cost	Applies To
PM Jan Dhan Yojana	Zero-balance bank account + ₹2L accident cover (RuPay)	All Indian citizens	Free	All personas
PMSBY (Suraksha Bima)	₹2L accident insurance	18–70 yrs, bank account	₹20/year	Ramesh, Suresh
PMJJBY (Jeevan Jyoti)	₹2L life insurance	18–50 yrs, bank account	₹436/year	All personas
PM Mudra Yojana	Collateral-free business loans up to ₹20L (Shishu, Kishore, Tarun, Tarun Plus)	Non-farm micro/small business owners	Interest set by lender	Suresh
Sukanya Samriddhi Yojana	Girl-child savings (8.2% p.a.)	Parents of a girl under 10 yrs	Min ₹250/year	Priya
Atal Pension Yojana	₹1,000–5,000/month guaranteed pension at 60	18–40 yrs, unorganized sector	₹42–1,454/month	Ramesh, Priya
PMEGP	Up to 35% subsidy on project cost for new micro-enterprises	18+ yrs, new non-farm enterprise	Free to apply	Suresh
PM-JAY (Ayushman Bharat)	₹5L health cover per family per year	Eligible low-income families (SECC-based)	Free	Ramesh, Priya
PM Fasal Bima Yojana	Crop insurance against yield loss	Farmers growing notified crops	1.5–5% of sum insured	Farmer persona

Scheme figures verified October 2026. Rates (e.g. Sukanya Samriddhi) are revised quarterly by the Government of India — see Risks (Section 23) for how SAHAI keeps this current.

19. Customer Service & Growth Strategy

SAHAI follows an AI-first customer support approach designed for users with varying financial needs and levels of literacy.

Tier 1: In-Product AI Support

- Conversational support through voice and text
- Hinglish-first interactions
- Instant responses for common financial queries
- Government scheme information and guidance

Tier 2: Human Escalation (Future Phase)

- Human support for unresolved issues
- Callback-based help for complex cases
- Local-language support

> **Goal:** accessible, low-friction support that feels like talking to a trusted financial companion rather than a traditional helpdesk.

19.1 Increasing CLV & Renewal Rates

User Engagement Strategy

- Personalized guidance based on the user profile
- Contextual government scheme recommendations
- WhatsApp nudges for reminders and habit formation
- Continuous value through recurring interactions

Retention Drivers

- Relevant, timely recommendations
- Consistent experience across channels
- Habit-building through proactive nudges
- Expanding scheme and financial knowledge coverage

B2B Retention (Future)

- Demonstrate user engagement metrics
- Showcase conversion and activation improvements
- Provide periodic performance reviews to partners

19.2 Growth Hacks & Growth Loops

1. WhatsApp Sharing Loop

Users share useful scheme recommendations and financial insights with friends and family on WhatsApp.

```
User receives value → Shares insight → New user joins → New engagement
```

2. Government Scheme Discovery Loop

Users discover relevant schemes through SAHAI and return periodically to check for new opportunities.

```
Scheme discovery → User benefit → Repeat usage → Higher engagement
```

3. Nudge-Based Retention Loop

Personalized WhatsApp reminders bring users back to interact regularly.

```
Nudge → User returns → Receives value → More engagement → Future nudges
```

4. Community Trust Loop

Users recommend SAHAI within their communities based on positive experiences.

```
Positive outcome → Word-of-mouth referral → New users → More trust
```

> **Goal:** drive organic growth through usefulness, trust and repeat engagement rather than paid acquisition.

20. Success Metrics

20.1 MVP Demo Metrics

Metric	Target / Measurement
Questions answered per session	≥3 complete voice Q&A cycles in the demo flow
STT accuracy on Hinglish	>90% correct transcription on scripted demo queries
End-to-end response latency	<5 seconds from mic release to audio playback
WhatsApp nudge delivery rate	100% delivery to the test recipient during the demo
Scheme recommendation relevance	3 relevant schemes surfaced per persona query

20.2 Adoption Metrics

Metric	Target
First-time voice query completion	>80%
Users asking a second question	>50%
WhatsApp opt-in rate	>70%
Weekly active users	>40%

20.3 Trust Metrics

Metric	Target
Advice helpfulness score	>4/5
Recommendation acceptance rate	>60%
User confidence improvement	>50%
Repeat utilization rate	>60%

20.4 Financial Impact Metrics

Metric	Target
Savings habit creation	>4/5
EMI reminders acted upon	>60%
Scheme awareness increase	>50%
Loan-risk avoidance	20%

21. Assumptions

1. Supply-Side Assumption: NBFC Partner Demand

Assumption: Regulated NBFCs will readily partner with us because they need to fill India's ₹28 lakh crore unorganized credit gap but lack the physical reach to do it safely.

Data: Banking surveys show that while 85% of smallholders have active bank accounts, less than 25% can access formal loans because of complicated branch paperwork. SAHAI addresses this by acting as a paperless, digital pipeline.

2. Demand-Side Assumption: The Institutional "Halo Effect"

Assumption: Low-income, unorganized segments will trust the app with sensitive financial details because we partner directly with licensed banks and their local field teams.

Data: Microfinance industry reports indicate that rural trust and onboarding conversion rise by over 3× when a digital application is co-branded with and introduced alongside a recognized financial institution.

3. User-Capability Assumption: Rural Digital Readiness

Assumption: Target users in Tier 2/3 regions are digitally capable enough to use a simple, voice-guided, single-OTP workflow.

Data: The IAMAI–Kantar report puts India at 958 million active internet users, with rural India driving 57% of that base; 44% of rural users already navigate apps using voice search and localized tools, or rely on shared family-device support.

22. Ethical Considerations

How our AI addresses the most basic financial problems responsibly.

Problem	Our Approach	Safety Guardrail
Bad credit records mistreat the poor	Traditional banks reject anyone without a credit history. Our AI looks at daily UPI sales or crop income to show someone can repay, instead of relying only on the credit bureau.	Following Dvara Research findings, we keep the scoring logic simple so a user isn't unfairly blocked for one bad month caused by rain or seasonal dips.
Complicated banking apps cause fear	We remove long, confusing English forms. A clear voice assistant guides the user step by step in their local language.	MicroSave Consulting notes that rural users get tricked by complex menus. We use simple voice check-ins: "Can we look at your sales to find a loan? Press Green for Yes."
Opaque AI gives wrong financial advice	The assistant is scoped, not a free-form chatbot: it answers from a pre-approved list of official government schemes and bank interest rates.	As EY and others warn, open-ended AI can hallucinate. For a small farmer or shopkeeper, a wrong interest rate can ruin a business — so we restrict answers to verified sources to minimise that risk.
Complex consent undermines autonomy	Forcing users to accept dense "Terms & Conditions" violates basic autonomy. FinDev Gateway (MicroSave Consulting) notes that standard disclosure is not consumer protection — design must reflect actual comprehension.	Align fully with the Digital Personal Data Protection (DPDP) Act. Consent is broken into modular steps with simple text or audio cues, e.g. "We are looking at your business data only to verify your income, not to read personal messages."

23. Risks & Limitations

Risk / Limitation	Description	Mitigation
Hardcoded scheme knowledge	LLM prompt contains static scheme info that may go out of date	Build a RAG pipeline in Phase 2 that pulls from government APIs or a scraper
Demo personas only	Real users have varied income/family structures not covered by 3 personas	Phase 2: dynamic onboarding flow to build the user profile
No real banking data	Advice is based on a hardcoded profile, not actual account data	Phase 4: Account Aggregator (AA) framework integration
Twilio sandbox limits	Sandbox WhatsApp only messages registered test numbers	Upgrade to Twilio production or Meta WhatsApp Cloud API post-demo
Sarvam free-tier limits	Free credits are enough for the demo, not for production volume	Switch to a paid plan or self-host Whisper for production
Groq rate limits	Free-tier request caps are adequate for the demo but would bottleneck at scale	Add request queuing + fallback to a second LLM provider (e.g. Gemini API)
LLM hallucination risk	LLM may give incorrect financial advice in edge cases	Add a disclaimer in the UI; scope queries to supported use cases
No user authentication	Demo runs on shared personas — not suitable for private data	Implement Supabase Auth in Phase 2

24. Future Roadmap

Phase	Timeline	Features	Goal
MVP	Complete	Voice AI, WhatsApp nudges and replies, 3 personas, Hinglish, scheme discovery	Demo-ready working product
Phase 2	0–3 months post-MVP	Dynamic onboarding, 5+ languages (Tamil, Marathi, Bengali...), user auth, RAG scheme database, scheduled nudges (APScheduler), production deployment	Real user onboarding at scale
Phase 3	3–6 months	Personalized savings goals, spending tracking and charts, financial health score, expanded scheme-eligibility engine, SMS support	Habit-forming financial companion
Phase 4	6–12 months	Account Aggregator (AA) integration, partner bank/NBFC APIs, credit health monitoring, micro-insurance recommendations, UPI-linked savings nudges	Full financial inclusion platform

25. MVP Screens & Demo

25.1 Demo Screens

All screens below are implemented and functional in the MVP. See the demo video in 25.2.

Screen	Description
Home Screen / Persona Selection	Persona selector (Ramesh / Priya / Suresh) and profile card showing name, occupation, income and spending summary. Saffron accent on a dark background, 390px mobile width.
Voice Interaction Screen	Mic button with recording timer and pulse animation, transcript box showing STT output, and the AI's Hinglish guidance.
Audio Playback	Waveform indicator during TTS playback, response card with SAHAI branding, replay button.
WhatsApp Nudge Panel	Nudge trigger button per persona, Hinglish nudge preview, and a green "Nudge Sent ✓" confirmation after dispatch.
WhatsApp Chat (Phone)	A real WhatsApp conversation on a test phone, showing the Twilio sandbox number (+1 415 523 8886) sending Hinglish financial advice and nudges.

25.2 Demo Links

Video Demo	drive.google.com/file/d/1qLRgxgZzJ9vMhSNBZgSpa7TE7wTMepOZ
GitHub Repository	github.com/harrysingh1908/sahai

26. Jira Board

Each functional requirement (FR-01 to FR-12) was tracked as a ticket on the SAHAI Kanban board.

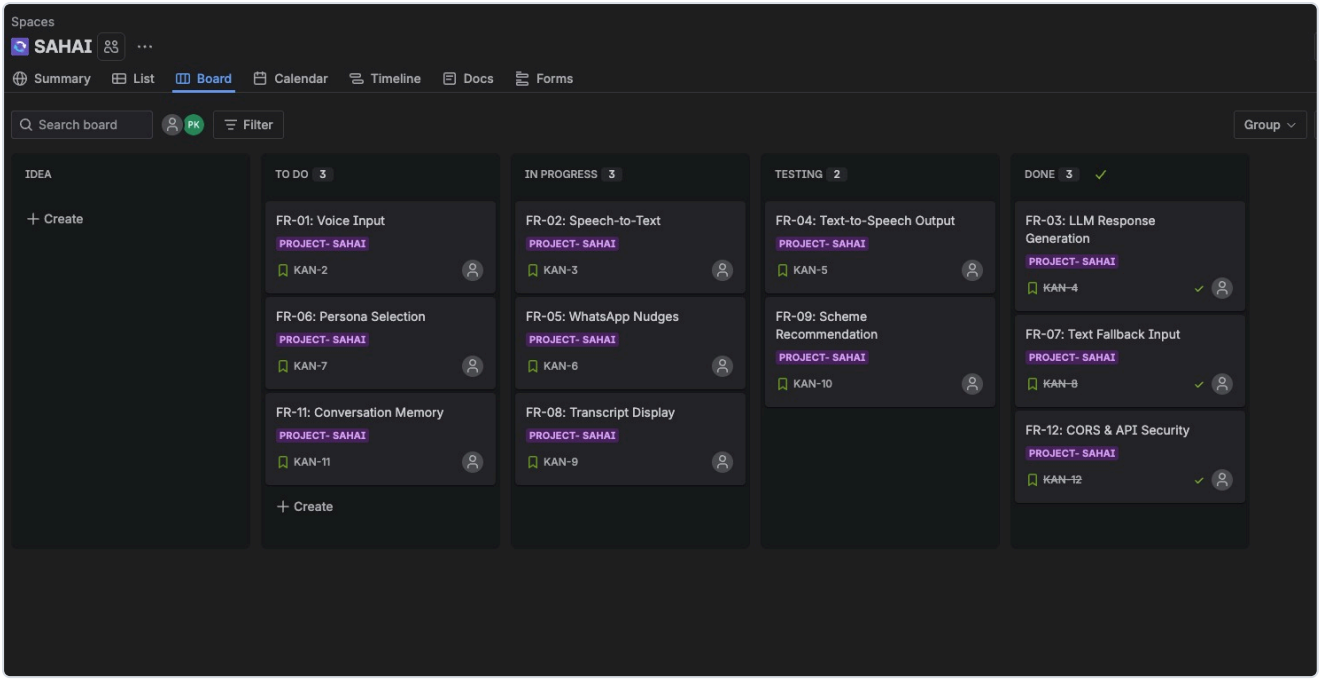


Figure 3 — SAHAI Jira board (FR tickets across Idea → To Do → In Progress → Testing → Done)

27. User Research — Field Interviews

Primary research was conducted through recorded interviews with people from the target segments.

Interviewee	Recording
Auto Driver	drive.google.com/file/d/1w2C9OLTnG9blsGaFqeOFMJ1mg3rIx7wv
Fast-Food Employee	drive.google.com/file/d/1qoJcfYebm2cKOuvR1p04nM4A9QibEVK0
Juice Seller (Part 1)	drive.google.com/file/d/1rxdZaaGuBmYinlOFLCjUIvOo73kfZ6NW
Juice Seller (Part 2)	drive.google.com/file/d/1wVDOBCUuRwYVe2xcpJzvgr5m2J3GdgwA
Business Owner (Seenema)	drive.google.com/file/d/1nVg-B-e6RF5pUEh2yUOCZGubJOX0r4Y8
Business Owner (Homelyfit)	drive.google.com/file/d/1PckDbb86zKZ_kwY0TViwibmpJQGrCvJC

SAHAI — सहाय

"Har faisle se pehle, SAHAI se poochho."

("Before every decision, ask SAHAI.")